The story of how black holes radiate and evaporate was derived by Hawking using the techniques of conventional quantum field theory, just in a curved spacetime of general relativity rather than a particle physicist's usual no-gravity context. It's not a genuinely quantum-gravity result; spacetime itself is treated classically, not as part of the quantum wave function. But nothing about the scenario actually seems to require deep knowledge of quantum gravity. As far as physicists can tell, Hawking radiation is a robust phenomenon. Whenever we do figure out quantum gravity, in other words, it should reproduce Hawking's result.

That raises a problem, one that has become notorious within theoretical physics as the *black hole information puzzle*. Remember that quantum mechanics, in its Many-Worlds version, is a deterministic theory. Randomness is only apparent, arising from self-locating uncertainty when the wave function branches and we don't know which branch we're on. But in Hawking's calculation, black-hole radiation seems not to be deterministic; it's truly random, even without any branching. Starting from a precise quantum state describing matter that collapses to make a black hole, there is no way of computing the precise quantum state of the radiation into which it evaporates. The information specifying the original state seems to be lost.

Imagine taking a book—maybe the very one you are reading right now—and throwing it into a fire, letting it burn completely away. (Don't worry, you can always buy more copies.) It might appear that the information contained in the book is lost in the flames. But if we turn on our physicist's powers of thought-experiment ingenuity, we realize that this loss is only apparent. In principle, if we captured every bit of light and heat and dust and ash from the fire, and had perfect knowledge of the laws of physics, we could reconstruct exactly what went into the fire, including all the words on the pages of the book. It'll never happen in the real world, but physics says it's conceivable.

Most physicists think that black holes should be just like that: throw a book in, and the information contained in its pages should be secretly encoded in the radiation that the black hole emits. But this is not what happens, according to Hawking's derivation of black-hole radiation; rather, the information in the book appears to be truly destroyed.

It's possible, of course, that this implication is correct, that the information really is destroyed, and that black-hole evaporation is nothing